

Empirical Design Notes: Sample Construction, Units, and Event Exclusions

Rate-Based Reference Model (Direction F)

Scope

This note documents the panel construction for the cross-sectional test of $\phi^- = 1 - (\text{ROE} - g)$: the units basis (Section 1.1, where an error in an earlier draft is corrected), what the regressor’s variance decomposition implies about what the test actually identifies (Section 2), and the principled grounds for excluding specific country-episodes (Section 5). Every exclusion is derived from a stated model assumption failing, not from an observation being inconvenient for the predicted slope.

1 Data

The buffer z (regulatory capital to risk-weighted assets, IMF FSI code `FSKRC_PT`) and the required return ρ (approximated by return on equity, `FSERE_PT`) are drawn from the IMF Financial Soundness Indicators dataset at quarterly frequency, Core Set, Deposit Takers. Growth g is drawn from the IMF World Economic Outlook database.

1.1 Units: the regressor must be computed on a nominal basis

FSI return on equity is reported in nominal, domestic-currency terms: it is net income (a nominal flow) divided by book equity (a nominal stock). Required capital grows with the nominal balance sheet, since risk-weighted assets are carried at nominal values. The regressor $\text{ROE} - g$ is therefore only correctly specified when g is *nominal* GDP growth in the same domestic currency — not real GDP growth.

An earlier version of this analysis used WEO’s `NGDP_RPCH` (real, constant-price growth) throughout. This was an error, not a stylistic choice, and its effect is not small. Recomputing g from WEO’s `NGDP` (current-price GDP level, domestic currency) as $g_t = \text{NGDP}_t / \text{NGDP}_{t-1} - 1$ and averaging over each country’s sample window changes the regressor materially:

	real-basis (wrong)	nominal-basis (correct)
mean of $\text{ROE} - g$	10.34pp	4.31pp
sd of $\text{ROE} - g$	5.12pp	7.14pp
range	−2.95 to 24.00	−34.42 to 16.53
countries with negative rebuild capacity	3	9

The starkest illustration is Argentina, whose regressor value moves from +24.00 (the largest positive value in the real-basis panel) to −34.42 (the largest negative value in the nominal-basis panel) — a 58-point swing driven entirely by which growth series is used, for one country. Turkey,

Belarus, Angola and Ghana show swings of 15–26 points for the same reason. The implied inflation proxy ($g_{\text{nominal}} - g_{\text{real}}$) exceeds 10 percentage points for seven of seventy-five countries and has a panel median of 4.2pp, so this is not confined to a handful of extreme cases; it is a first-order correction to the regressor for a meaningful share of the sample.

All results in this note and in any subsequent estimation use the nominal basis. The real-basis numbers should not be reported, including as a robustness check, since they are not an alternative specification of the same question — they are the wrong quantity for the model as written.

2 What the regressor’s variance decomposition implies

Across the panel (nominal basis, $n = 75$), ROE has standard deviation 4.95pp and g has standard deviation 1.62pp (real) or substantially more on a nominal basis in the high-inflation members; the correlation between ROE and real g is 0.06. Decomposing the regressor’s variance, $\text{Var}(\text{ROE} - g) = \text{Var}(\text{ROE}) + \text{Var}(g) - 2 \text{Cov}(\text{ROE}, g)$, cross-country variation in ROE accounts for the large majority of the regressor’s spread on the (mis-specified) real basis; on the nominal basis, variation in g itself is substantially larger, since nominal growth ranges from near zero to over 50% across the panel’s high-inflation members.

Two consequences follow.

First, whichever basis is used, the paper’s identifying variation is not concentrated in near-zero-growth, low-inflation economies of the kind (Japan, stagnant Western Europe) that motivate the theory narratively. That mismatch between motivating case and estimating sample must be stated explicitly rather than left for a reader to notice.

Second, and more consequential on the nominal basis: the regressor now spans very high nominal growth rates in chronically high-inflation economies (Argentina, Turkey). Whether the linear model $\phi^- = 1 - (\text{ROE} - g)$ is a sensible description of buffer dynamics in a hyperinflationary or near-hyperinflationary economy is not established by anything in this project and should not be assumed; these observations may need their own treatment (Section 5) rather than being pooled with the rest of the panel at face value.

2.1 Negative rebuild capacity

Nine countries (nominal basis) have $\text{ROE} < g$, implying $\phi^- > 1$: an explosive buffer under the model’s linearisation. This is economically a coherent prediction — a banking system that cannot fund its own required growth from earnings needs continuous external capital injection to avoid a widening deficit — but it is not estimable as an ordinary autoregressive coefficient, and four of the nine (Argentina, Turkey, Belarus, Tajikistan) are also flagged above for basis-sensitivity.

Decision, fixed before estimation (31 July 2026). All nine are treated as a **separate documented subsample**, reported alongside the main estimation rather than pooled into it or dropped. Two of the three options considered are rejected on stated grounds: an estimator accommodating $\phi^- \geq 1$ is not adopted, because no such estimator has been validated on known truth in this project — standing practice throughout this project is to validate any estimator against simulated data of known truth before applying it to real data — and adopting one now, for exactly the observations that would otherwise embarrass a linear fit, is the kind of choice this document exists to prevent; outright exclusion under Section 5 is not adopted either, because negative rebuild capacity is not one of the four violations of B1–B2 listed there — it is a prediction of the model at these parameter

values, not a breakdown of its premises, and excluding a model’s own predicted regime because it is awkward to estimate is not a defensible exclusion ground. A separate subsample is the only option that reports the observation rather than discarding or mis-estimating it. The main-sample result and the nine-country subsample result must both be reported; neither may be presented alone.

3 The deleveraging channel: B2 is incomplete, but the omission is not growth-correlated

Assumption B2 states that a capital deficit can only be closed by earning it back. That is incomplete: a deficit can also be closed by shrinking the denominator — shedding risk-weighted assets. Post-2008 European bank capital ratios recovered substantially through balance-sheet contraction rather than through profitability, so this is not a hypothetical channel.

The concern this raised was sharper than mere incompleteness. If deleveraging is *cheaper when growth is low* — shedding assets costs market share in a boom and much less when every competitor is shrinking — then the omitted channel would be correlated with g , and the regression of ϕ^- on $(\text{ROE} - g)$ would suffer omitted-variable bias in a known direction: steeper than -1 . That is the same direction as the small-sample AR bias already documented, so the two would be confounded and neither could be identified from the other.

Test. Using IMF FSI risk-weighted-asset levels (FS_ODX_ARW_FSKRC_XDC, available for 66 of the 77 panel countries with clean overlap), quarterly RWA growth was computed and split by regime, with the deficit indicator taken from the buffer ratio. The statistic is the gap: surplus-quarter RWA growth minus deficit-quarter RWA growth, so a positive gap means RWA grows more slowly (or contracts) when the buffer is under pressure.

Result 1: the channel is real. Sixty-four percent of countries show a positive gap, with a cross-country mean of 1.19 percentage points per quarter. Banks do shrink risk-weighted assets when capital is short. B2 as written omits something that happens.

Result 2: it is not growth-dependent, and the identification concern does not survive.

The correlation between a country’s average nominal growth and its deleveraging gap is $+0.33$ — the *opposite* sign to the one that would generate the feared bias. That weak positive association is itself fragile: it falls to $+0.22$ once Eswatini and Argentina are excluded (both already flagged in Section 5; Eswatini’s gap of 19.7 is more than double any other country’s, and Argentina’s nominal growth is almost entirely inflation), and the rank correlation — which prevents two extreme observations from driving the estimate — is 0.19 ($p = 0.13$) on the full sample and 0.16 ($p = 0.20$) excluding those two. A median split on growth is the clearest statement: low-growth countries average a gap of 1.17, high-growth countries 1.21. Indistinguishable.

Consequence. The deleveraging channel is a **model-completeness** issue, not a **slope-identification** threat. It belongs in the theory as an extension — a deficit has a slow cheap route (earn) and a fast costly one (deleverage), with the mix depending on how binding the shortfall is — but it does not need to be resolved before the main cross-sectional estimation can be trusted, and the overshoot to $-1.2/ -1.3$ observed in simulation remains attributable to small-sample AR bias rather than to this omission. This corrects a concern raised earlier in the project’s development, which anticipated the opposite finding.

Coverage note. Ten panel countries have no RWA level series at all: Ecuador, Estonia, Croatia, Latvia, Malta, Netherlands, Palestine, Slovakia, **the United States**, and Kosovo. This is a coverage gap on the outcome variable, layered on the gaps already documented for the regime variables, and the absence of the United States in particular should be stated rather than passed over.

Eswatini. The largest single outlier (19.7pp gap, more than twice the next) was investigated rather than left unexplained. No documented systemic banking crisis or bank resolution was found. What does exist is a well-documented 2010–11 fiscal crisis — a collapse in Southern African Customs Union revenue — at the very start of Eswatini’s FSI window, combined with an IMF assessment that Eswatini’s banks carry large exposures to the government such that a fiscal shock propagates directly into the financial sector. This is a sovereign shock transmitted to a very small banking system, not a banking crisis of the kind Section 5 enumerates. It is therefore a candidate for the size-based screen rather than the event-based one, and is flagged for manual review rather than automatic exclusion.

4 Estimating specification: regional fixed effects, with pooled OLS reported alongside

Comparing a small sub-Saharan banking system directly against a large European one attributes to the model’s mechanism what may simply be structural difference between economies operating under different pressures. The main specification therefore includes regional fixed effects, identifying the slope from *within*-region variation in $(ROE - g)$:

$$\hat{\phi}_i^- = \alpha_{r(i)} + \beta (ROE_i - g_i) + \varepsilon_i, \quad \text{model prediction } \beta = -1.$$

Pooled OLS without regional effects is reported alongside, so a reader can see directly whether the result depends on cross-regional comparison.

Grouping. Regions are assigned on **geographic proximity alone**. Political affiliation, trade arrangements, currency blocs and alliance structures are deliberately ignored; the grouping is meant to capture the shared economic environment that comes with location, not institutional membership. Eleven regions result:

Region	n	mean $(ROE - g)$	within-region sd
Europe	19	3.02	4.22
Sub-Saharan Africa	16	6.69	5.47
Latin America	14	3.15	11.23
Southeast Asia	6	4.93	3.02
Oceania	5	10.88	4.82
West Asia	3	−2.25	6.35
South Asia	3	0.99	7.94
East Asia	3	6.65	3.48
Middle East	2	7.31	0.82
North America	2	8.78	5.37
Central Asia	2	−8.81	4.87

What fixed effects cost. Decomposing the regressor’s variance, 24.9% is between-region and 75.1% within. Regional effects therefore discard about a quarter of the identifying variation, leaving three quarters — ample for the slope test — at a cost of eleven degrees of freedom ($n = 75$ falls to 63 residual df). No region is a singleton, so every observation contributes.

Three grouping decisions, stated rather than buried. Georgia, Armenia and Turkey are land-contiguous with both Europe and the Middle East; rather than force a choice they are grouped as West Asia. Korea is placed with Hong Kong and Macao as East Asia, not Southeast Asia. Bhutan, Sri Lanka and the Maldives form South Asia; the region is not labelled “India” because India is not in the panel. Each of these would shift the relevant regional means, and any of them is arguable; they are recorded here so the choice is auditable rather than implicit.

An incidental benefit, and a cost. Geographic grouping absorbs some concerns that had been handled by separate rules. Latin America’s within-region standard deviation of 11.23 — far the largest, and driven mostly by Argentina — means regional effects partially absorb the high-inflation problem of Section 1.1 without a dedicated exclusion. Central Asia separating cleanly from the Middle East (-8.81 against $+7.31$) is the grouping working as intended: an earlier ad-hoc grouping had averaged those together and concealed the difference.

The cost is specific and should not be glossed. With between-region variation removed, the specification can no longer test whether structurally slower-growing *regions* carry systematically higher persistence. Given the growth distribution documented in Section 2 — the panel contains almost no genuine stagnation cases — little is lost in practice, but the cross-regional form of the stagnation claim becomes untestable by construction rather than merely underpowered. That is why pooled OLS is reported alongside rather than discarded.

5 Economic grounds for excluding specific episodes

Assumptions B1–B2 describe a going concern whose capital adjusts only through retained earnings (deficit) or a payout decision (surplus). Four situations are identifiable, in advance, as violations of this description rather than as instances of it:

1. **External recapitalisation.** Where a shortfall is closed by a public injection, a bailout facility, or a state guarantee converted to equity, the deficit closes by fiat, not by earning it back. This is the single most consequential exclusion, because it is *not neutral*: a recapitalised deficit closes fast without being earned, which pushes the observed $\hat{\phi}^-$ down. Removing these episodes therefore *raises* estimated persistence and flatters the model’s prediction at low growth — the opposite of a conservative adjustment, and it must be reported as such.
2. **Payout restriction.** Where dividends and buybacks are prohibited by supervisory action — the near-universal 2020–21 restriction across advanced-economy banking systems being the largest instance in this sample window — δ_p is set to zero externally, so $\phi^+ = 1$: surpluses become perfectly persistent, exactly the reverse of B2’s assumption that a surplus is a policy choice.
3. **Resolution or nationalisation.** Where a bank is resolved, nationalised, or subject to an imposed balance-sheet contraction, the going-concern premise behind B1’s growth identity does not hold.

4. **Definitional discontinuity.** The Basel II to Basel III transition changed what counts as regulatory capital, and adoption dates differ by jurisdiction, producing a level break in z at a country-specific date unrelated to any economic event.

A fifth, added after the units correction: chronic high inflation. Where a country’s nominal growth rate is dominated by inflation rather than real activity (Argentina and, over parts of the sample window, Turkey and Belarus), the linear relationship $\phi^- = 1 - (\text{ROE} - g)$ conflates a real economic mechanism with a nominal accounting phenomenon that the model does not describe. These are candidates for a separate high-inflation subsample rather than for the main estimation, pending a decision on whether the model should be respecified in real terms with a correspondingly real measure of ROE.

Protocol. Event windows and the high-inflation flag are fixed from external, pre-existing sources — Laeven and Valencia’s systemic banking crisis database, in its current vintage extending through 2025 (IMF Working Paper WP/26/94, May 2026), for the first four grounds; a simple threshold on average CPI inflation, fixed before estimation, for the fifth — *before* any regression is run. Results are reported with the flagged episodes and countries included and excluded, side by side. A rule chosen after seeing its effect on the estimated slope is not a rule.

What this is not. The country-level amplitude screen used earlier in sample construction (excluding, on the basis of return-on-equity volatility alone, observations such as Ukraine during the war, the 2013 Cyprus bail-in, and the Greek banking crisis) conflates measurement noise from a national ratio dominated by one or two institutions with genuine systemic stress. The former is a data-quality problem and a candidate for exclusion; the latter is exactly the kind of episode a model of capital-buffer dynamics under stress ought to be tested against, not screened away from. Where volatility is high for an identifiable systemic reason, the preferred treatment is a robust estimator that limits its leverage, or an explicit crisis subsample reported alongside the main result, not deletion.

6 The country universe

Earlier versions of this panel drew on a fixed list of 77 areas. That list excluded roughly half of the areas reporting the capital ratio, including several with more usable history than areas it retained, and no rationale for it is recorded anywhere in this project. Two candidate explanations were tested directly against the source and both were rejected: it is not the set of areas reporting both the capital ratio and return on equity (those two indicators are reported by an identical set of areas), and it does not correspond to any standard advanced-versus-emerging classification. It is not reconstructed here. It is replaced.

The rule. The universe is every area the source reports for the indicator that is a sovereign state or dependent territory — operationally, an ISO 3166-1 alpha-2 code, together with Kosovo, which is an exceptional reservation rather than a formally assigned code and is admitted by name. Codes that are regional or income-group aggregates rather than reporting jurisdictions are excluded and named individually in the panel output. The rule is implemented once, in `panel_universe.py`, and imported by every script that needs a country universe, so the two panel constructions cannot drift apart.

No history screen at this stage. Whether an area has enough observations to support an estimate is a property of the estimator, not of the universe, and it is applied inside the estimation loop and reported as attrition alongside every other reason an area drops out. Folding a feasibility threshold into universe construction is what made the previous list unauditable: it merged the question of which areas exist for this purpose with the question of which areas carry enough data, and left only the intersection visible. Keeping them separate means every exclusion is attributable to a stated cause.

7 Currency-union pseudo-replication: a checked and rejected exclusion

When the country universe was rebuilt (Section 6), two clusters of areas were noticed to return near-identical numbers of observations: the Eastern Caribbean-adjacent group (Anguilla, Antigua and Barbuda, Dominica, Grenada, St Kitts and Nevis, St Lucia, Montserrat, St Vincent and the Grenadines) and the CEMAC-adjacent group (Cameroon, Central African Republic, Republic of Congo, Gabon, Equatorial Guinea, Chad). Both are currency unions with shared supervisory arrangements, so the natural suspicion was that their members report a single aggregated banking system, in which case treating them as independent observations would insert eight or six near-duplicate points into any cross-country median or dispersion statistic.

Decision, taken before testing. The suspicion was recorded as a flag in the panel output, not acted on as an exclusion. A pattern in observation counts is not evidence about the underlying series, and excluding countries on a plausible-sounding conjecture is precisely the practice this document exists to prevent.

Test. `panel_currency_union_check.py` compares the actual reported value series, pair by pair within each cluster: the correlation across jointly observed quarters, and the fraction of quarters where two members report numerically identical values. Shared reporting would produce correlations near one together with a high identical-value fraction.

Result: the suspicion is wrong. Across all 28 Eastern Caribbean pairs and all 15 CEMAC pairs, the fraction of numerically identical quarters is *zero*. Correlations are scattered across roughly $[-0.77, +0.74]$, many of them negative, with no pair approaching the near-unity that duplicated reporting would produce. The single highest, Equatorial Guinea and Chad at $+0.74$ with no identical values, is unremarkable co-movement between two small oil-exporting economies sharing a currency and a commodity cycle — an economic correlation, not a reporting artefact.

Consequence. These are independent national series whose sample windows happen to begin together, which is what a common supervisory reporting start date produces. No exclusion is warranted, the flag has been removed from the panel script, and this section is the record of why. The test should be re-run if the source's coverage changes.

8 The Basel III attempt: a documented negative result

Applying ground 4 to real data did not go as the diagnosis leading to it suggested it would, and the failure is instructive enough to record in full rather than pass over.

The pattern that motivated the attempt. Constructing z from each country’s own expanding-mean reference (Section 1.1 and its validation) and applying it to the real 77-country panel produced only 26 non-degenerate country estimates out of 77 — a two-thirds attrition rate against the classification guard. Among the thirteen countries with a *complete* surplus skew (zero deficit quarters observed at all), every single one had an FSI sample window spanning the original Basel III capital conservation buffer phase-in, sourced to the Bank for International Settlements’ own international timetable (1 January 2013 for the CET1 step-up, phased to full effect 1 January 2019) and confirmed directly for two of the thirteen: South Africa, via the South African Reserve Bank’s own Banks Act Circular 4 of 2016, and Indonesia, via OJK Regulation No. 11/POJK.03/2016. Eleven of the thirteen are BCBS, EU, or G20 members, jurisdictions with essentially no discretion over whether to follow the international schedule.

The mechanism, confirmed by simulation before touching real data again. A one-time, Basel-style level shift injected into an otherwise clean simulated capital-buffer process reproduces the symptom: degeneracy under a continuous expanding-mean reference rises from near zero to 40%. Segmenting the reference at the shift — restarting the expanding mean after the break rather than letting one continuous average straddle a change in what “normal capital” means — cuts that to 22%. The mechanism is real.

The result on real data. Segmenting the real panel’s expanding-mean references at 1 January 2019 did not improve the usable count: 25 non-degenerate countries against 26–27 under the unsegmented construction, a wash within noise. More importantly, *none of the thirteen targeted countries were rescued into non-degenerate status*. Two came close — Latvia and the Netherlands were four and three observations short of the minimum-regime threshold respectively — but most were not: Australia showed zero deficit quarters even measured against a reference that restarts after the transition, over roughly nine years of subsequent data. Segmentation instead lost five previously usable countries (Sri Lanka, Macau, Nicaragua, Eswatini, Türkiye) while gaining three different ones, one of which — the United States, with an anomalously large $\hat{\lambda} = 1.61$ — is better explained as noise from a thin post-break window than as signal.

Why this happened, and what it establishes. Segmenting a reference trades access for precision: splitting one long series into two shorter ones repairs the classification (the split becomes balanced) but shortens the sample each regime estimate draws on, and the resulting small-sample attenuation offsets the gain almost exactly. Simulation confirmed this in advance — mean $\hat{\lambda}$ under segmentation was statistically indistinguishable from the unsegmented value, both well below the true parameter used to generate the data — so the real-data outcome is not a surprise in hindsight, though it was not obvious before the attempt was made. A single calendar break, however well sourced, cannot simultaneously fit every country’s true contamination structure: it resolves the mechanism for some observations and introduces a new mismatch for others, netting out to no improvement in aggregate even where the underlying diagnosis is correct.

Disposition. Basel III timing is not abandoned as an explanation — it remains the best-supported account of why a large block of otherwise well-constructed observations classify as degenerate — but time-series segmentation is not the right tool to act on it. The recommended path, taken up after the remaining exclusion grounds in this section are applied, is to control for the transition at the *regression* stage: a post-2019 indicator or interaction term on the cross-sectional estimates from the *unsegmented* construction, which does not cost the sample length that segmentation does.

Whether that control absorbs the pattern better than segmentation did is not yet known and should not be assumed.